

# Operating System Last-Minute Revision Sheet

---

## 1. What is an Operating System?

- **Definition:** A program that acts as a bridge between the **user** and the **hardware** (e.g., laptops, smartphones).
  - **Role:** Manages resources like CPU, memory, storage, I/O devices, and files.
- 

## 2. OS Functions

- **Resource Management:** CPU, memory, I/O devices, storage.
- **System Boot:**
  - Bootstrap program starts.
  - OS kernel loads into memory.
  - Interrupts handle hardware/software events.

### 3. System Components

- Hardware: CPU, memory, I/O devices.
  - OS: Coordinates hardware and provides user interface.
  - Applications: Web browsers, games, databases.
  - Users: Humans or machines interacting with the system.
- 

### 4. Memory & Storage

- Execution Cycle: Fetch → Decode → Execute → Store result.
  - Memory:
    - RAM: Temporary, volatile, fast.
    - ROM/EEPROM: Static, non-volatile.
    - Storage Hierarchy: Registers → Cache → RAM → Disk → Tape.
- 

### 5. I/O Structure

- **I/O Devices:** Managed by **device controllers** and **drivers**.
  - **DMA:** Direct Memory Access reduces CPU load for large data transfers.
- 



## 6. Types of Operating Systems

- **Single User:** One user at a time (e.g., early PCs, mobile devices).
    - **Single Task:** One app at a time.
    - **Multiple Tasks:** Multiple apps for one user (e.g., multi-tasking mobile phones).
  - **Multiprogrammed System:** Runs multiple programs simultaneously.
  - **Batch Processing:** Jobs processed in a batch without user interaction.
  - **Time Sharing:** CPU switches quickly between jobs, offering each process a time slice.
  - **Real-Time System:** Immediate processing (e.g., industrial machines, medical systems).
-



## 7. OS Management Functions

- **Memory Management:** Tracks which processes use which parts of RAM.
  - **Processor Management:** Monitors CPU and manages process execution.
  - **Device Management:** Manages I/O devices (allocates/deallocates).
  - **File Management:** Manages files and access permissions.
  - **Security:** Protects data and resources (e.g., passwords).
- 



## 8. OS Services

- **User Interface:**
  - **CLI:** Command-Line Interface.
  - **GUI:** Graphical User Interface.
  - **Batch Interface:** Commands written in a file.
- **Program Execution:** Loads and runs programs.
- **I/O Operations:** Manages hardware access (e.g., printers).
- **File System Manipulation:** Create, delete, read, write files.

- **Communication:** Shared memory or message passing between programs.
  - **Error Detection:** Checks for hardware/software issues and takes actions.
- 



## **9. Key OS Functions**

- **Resource Allocation:** Fair distribution of CPU, memory, storage.
- **Accounting:** Logs resource usage for billing or analysis.
- **Protection and Security:** Prevents unauthorized access, protects against threats.